Autonomous Tractor

ECE 2804

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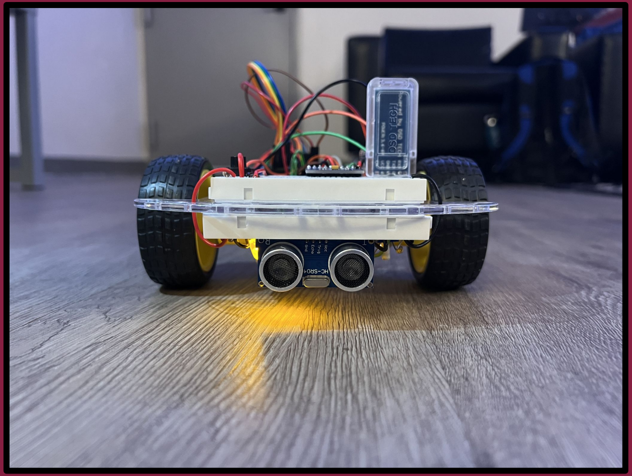
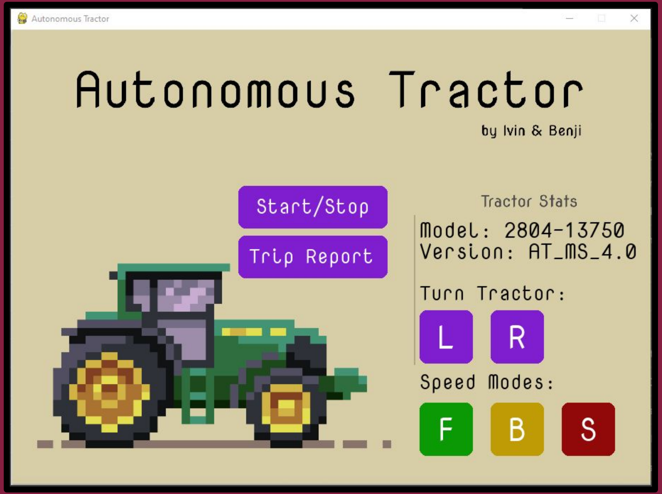
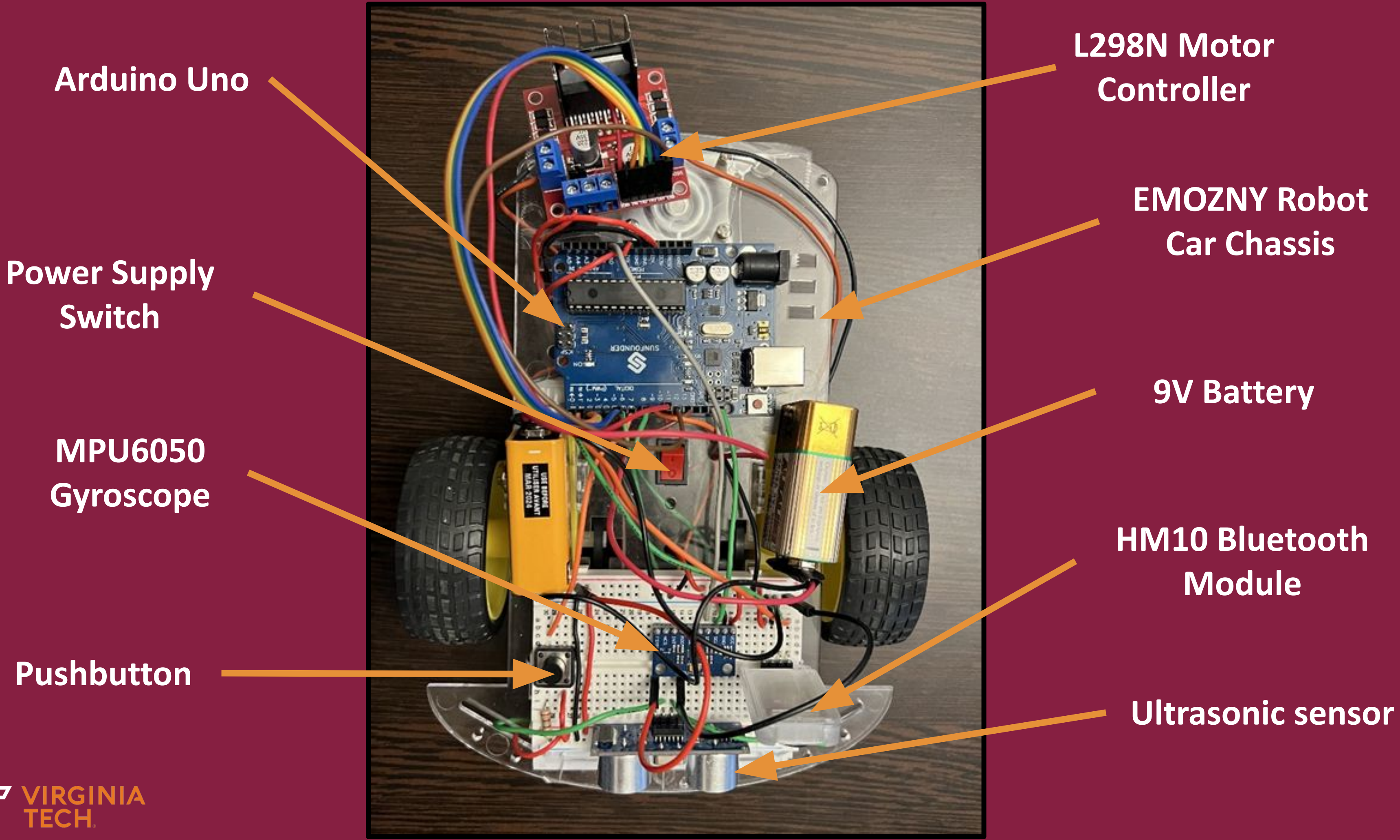
Introduction

GOAL:

Develop a tractor with the capability to drive straight, make turns and make emergency stops autonomously

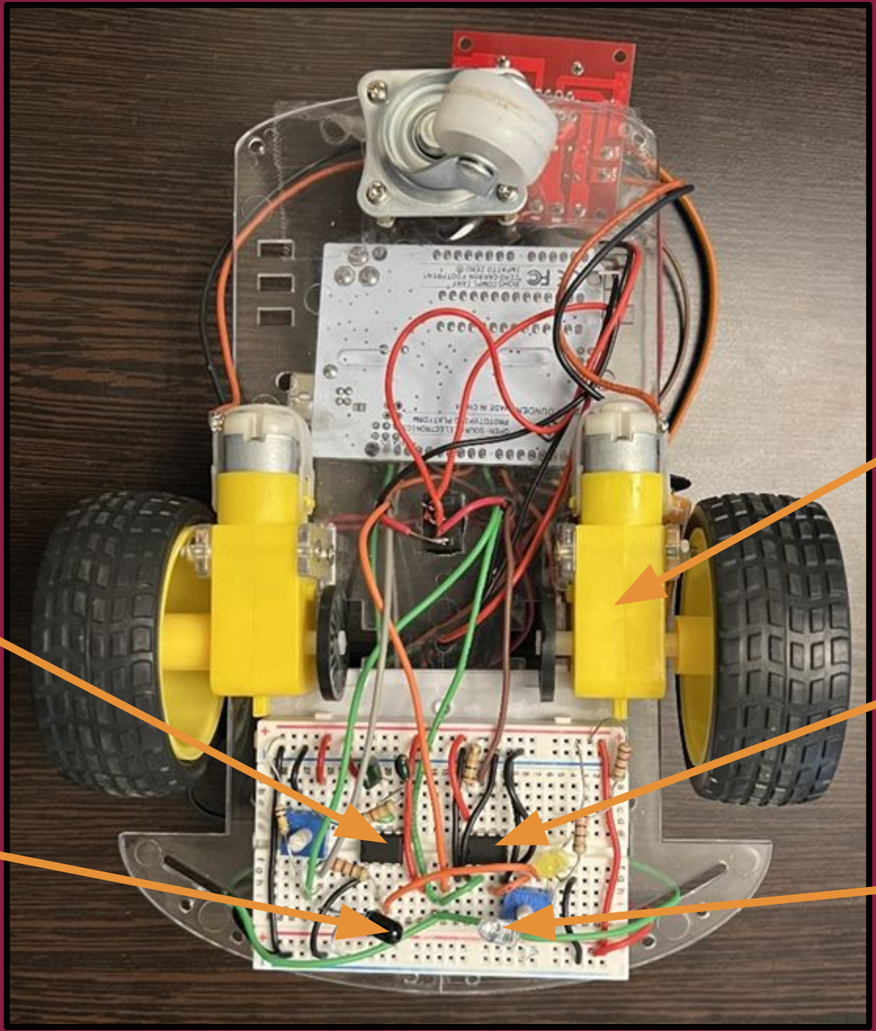





Arduino Uno
Power Supply Switch
MPU6050 Gyroscope
Pushbutton
L298N Motor Controller
EMOZNY Robot Car Chassis
9V Battery
HM10 Bluetooth Module
Ultrasonic sensor

555 Timer
(Hardware Solution)

Infrared Receiver
Diode



DC Motor

Comparator
(Hardware Solution)

Infrared Emitter
Diode

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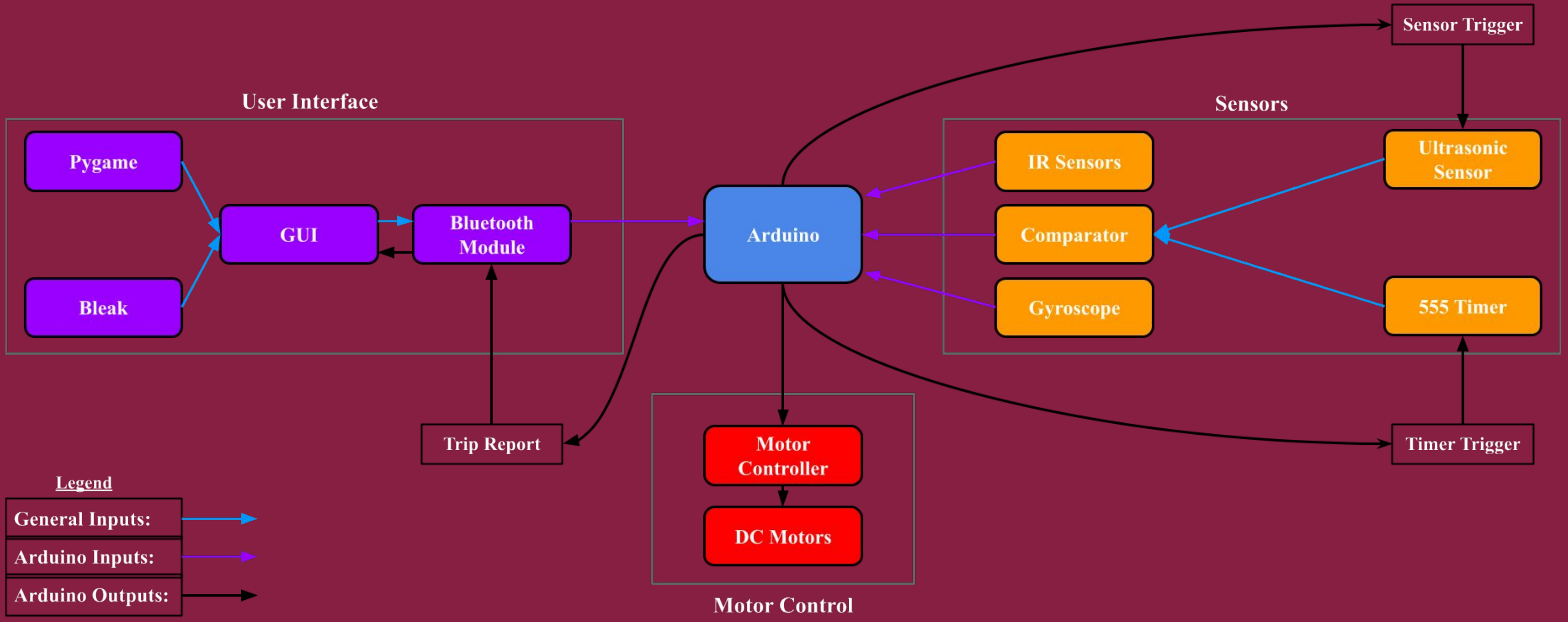
High Level Architecture

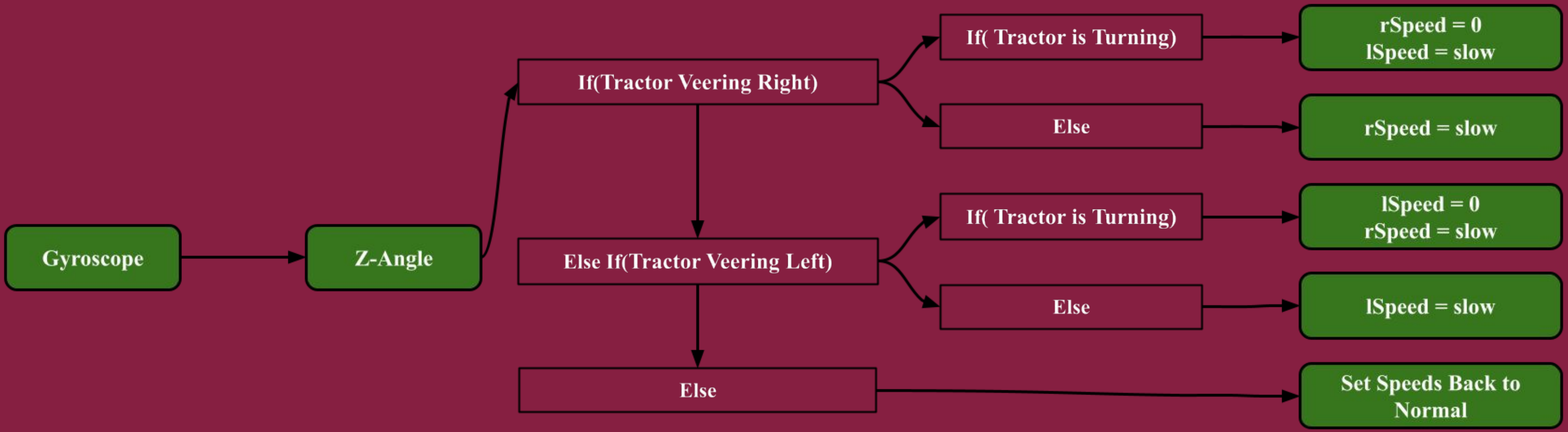


Arduino Code

```
void loop()
{
  straight(); // Uses gyroscope to drive the tractor straight
  bluetoothInterface(); // Uses HM10 to recieve commands from GUI
  pushButtonInterface(); // Toggles tractor if button is pressed
  turn(); // Uses IR sensors to tractor if black tape is detected
  eBrake(); // Uses Ultrasonic Sensor to E-Brake if obstruction deteccted
  initializeTractor(drive, rSpeed, lSpeed); // Sets the speeds and drive state
}
```

pushButtonInterface()

straight()



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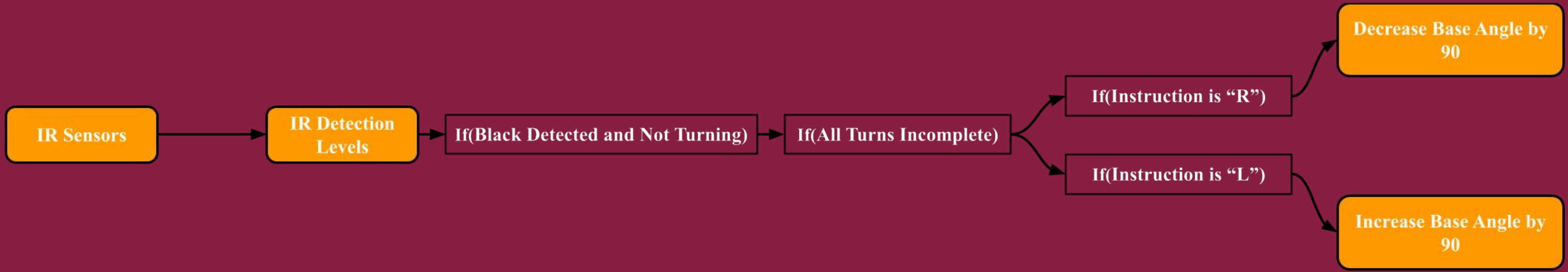
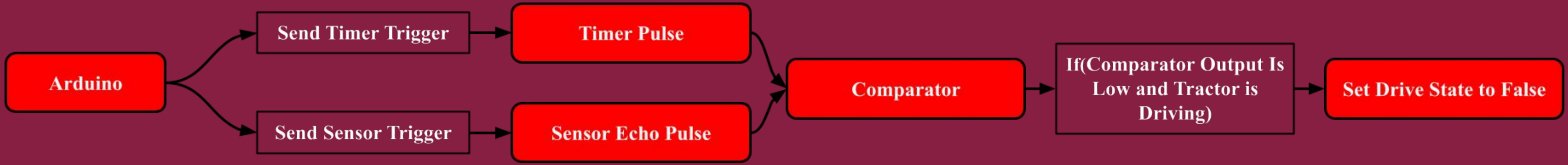
turn()


IR Sensors
IR Detection Levels
If(Black Detected and Not Turning)
If(All Turns Incomplete)
If(Instruction is "R")
If(Instruction is "L")
Decrease Base Angle by 90
Increase Base Angle by 90

eBrake()



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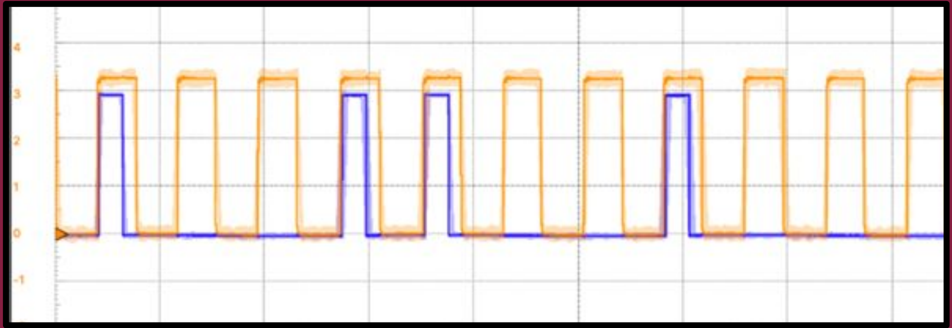
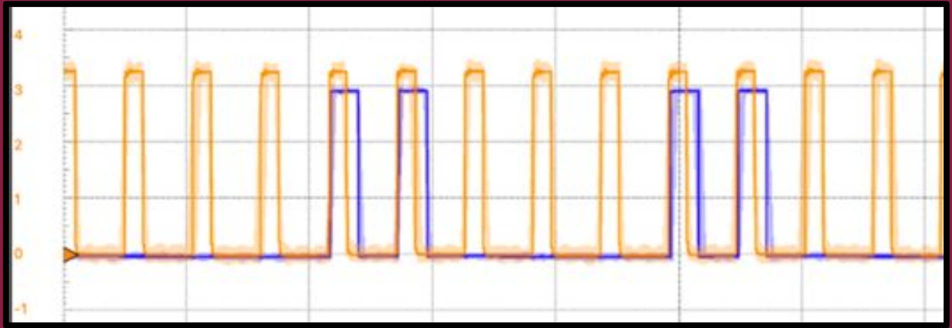
Hardware Solution Explanation



E-Brake Not Initiated



E-Brake Initiated

	Echo Pulse
	Timer Pulse

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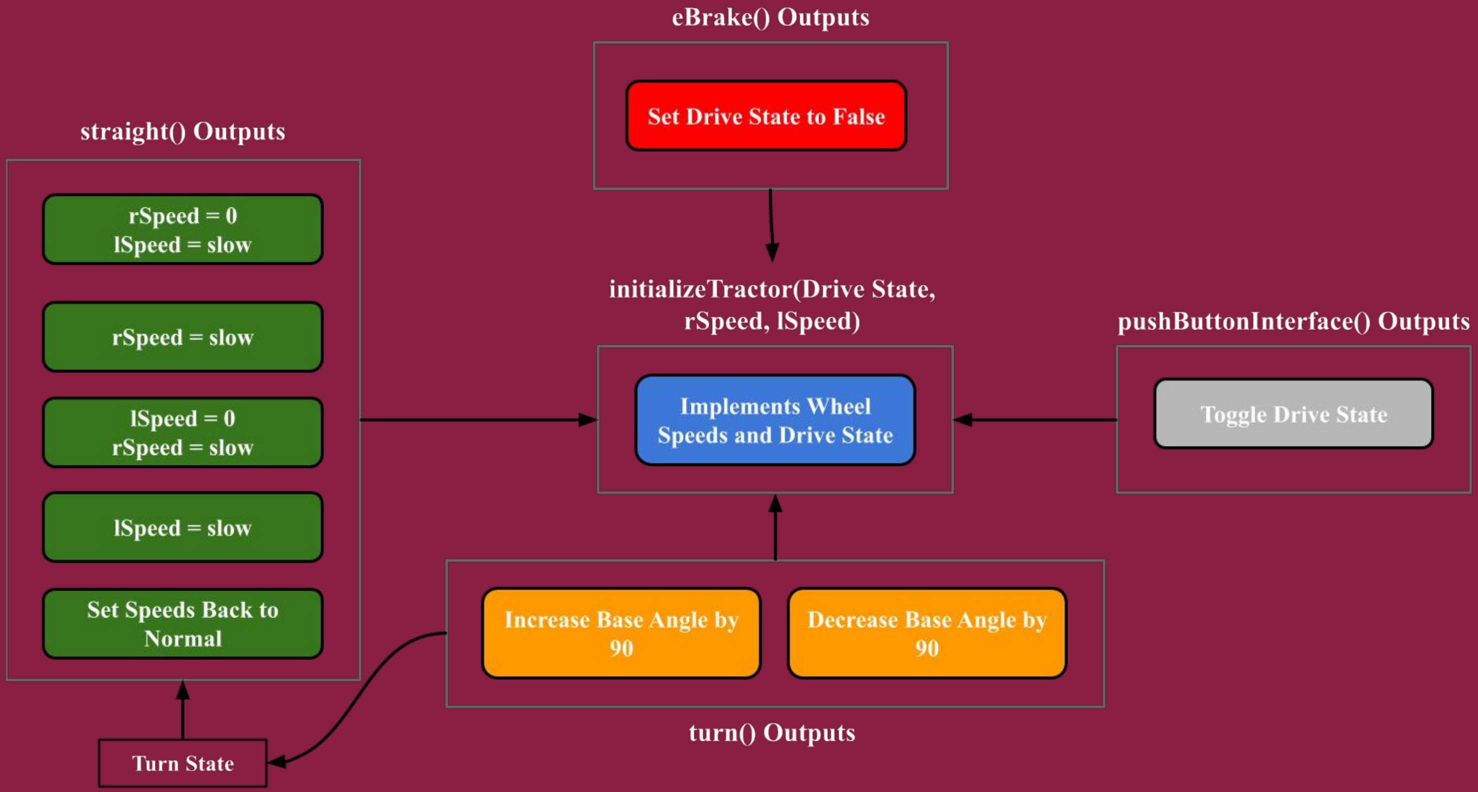
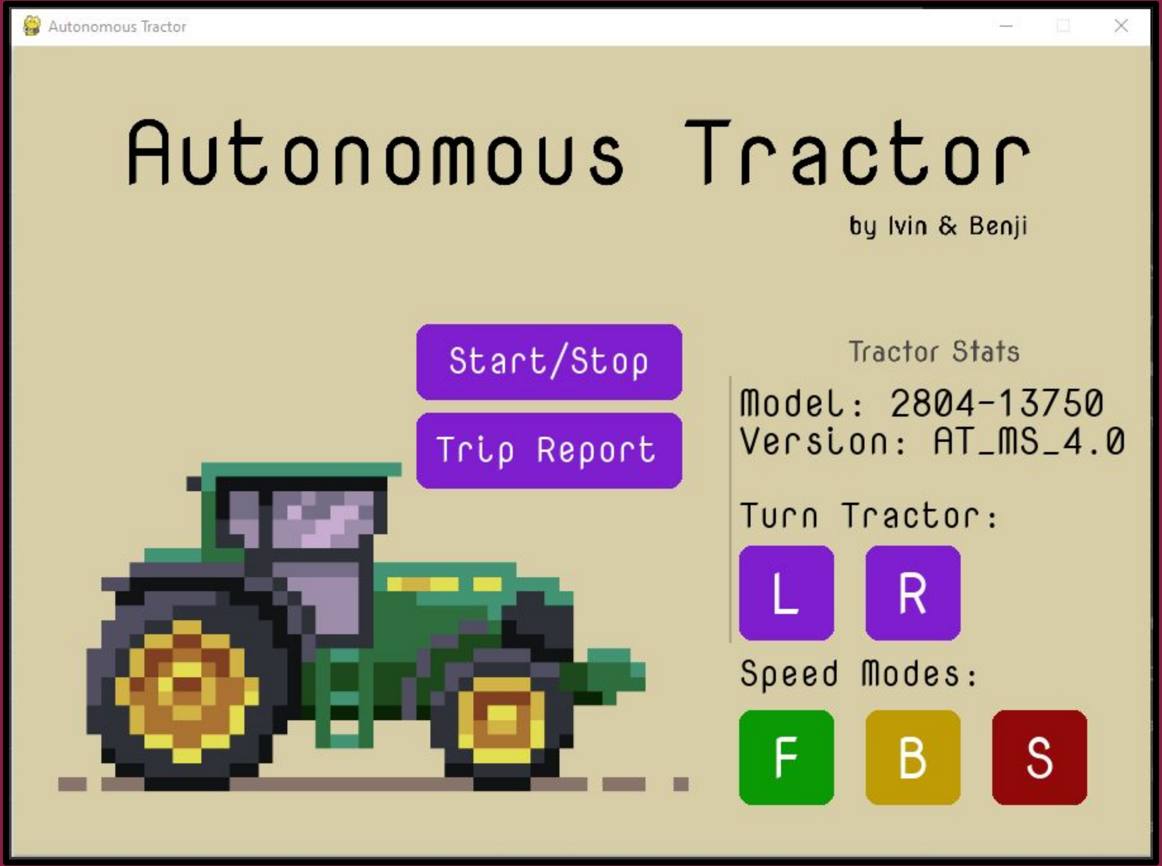
initializeTractor()

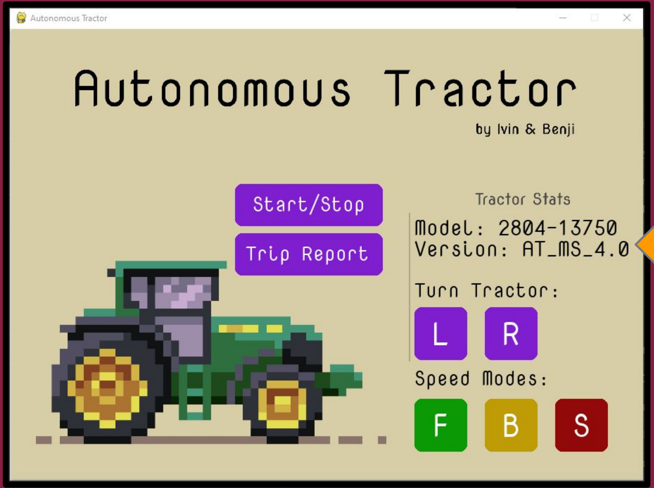


Graphical User Interface

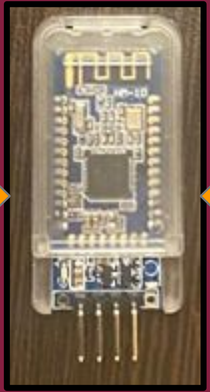
Designed using Figma!!!



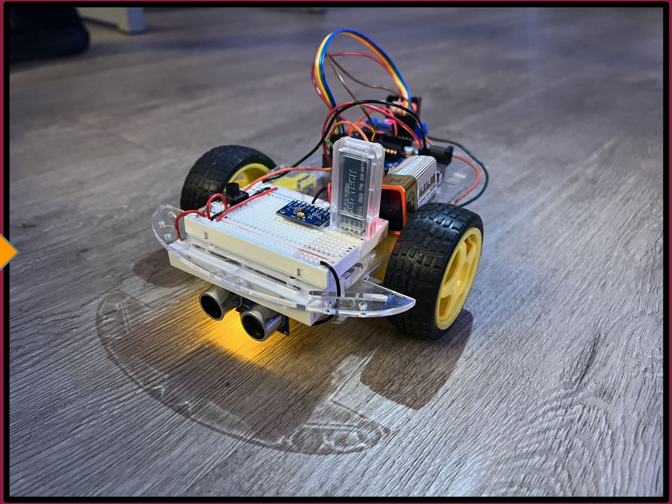
How it works…



Graphical User Interface



HM10 Bluetooth Module



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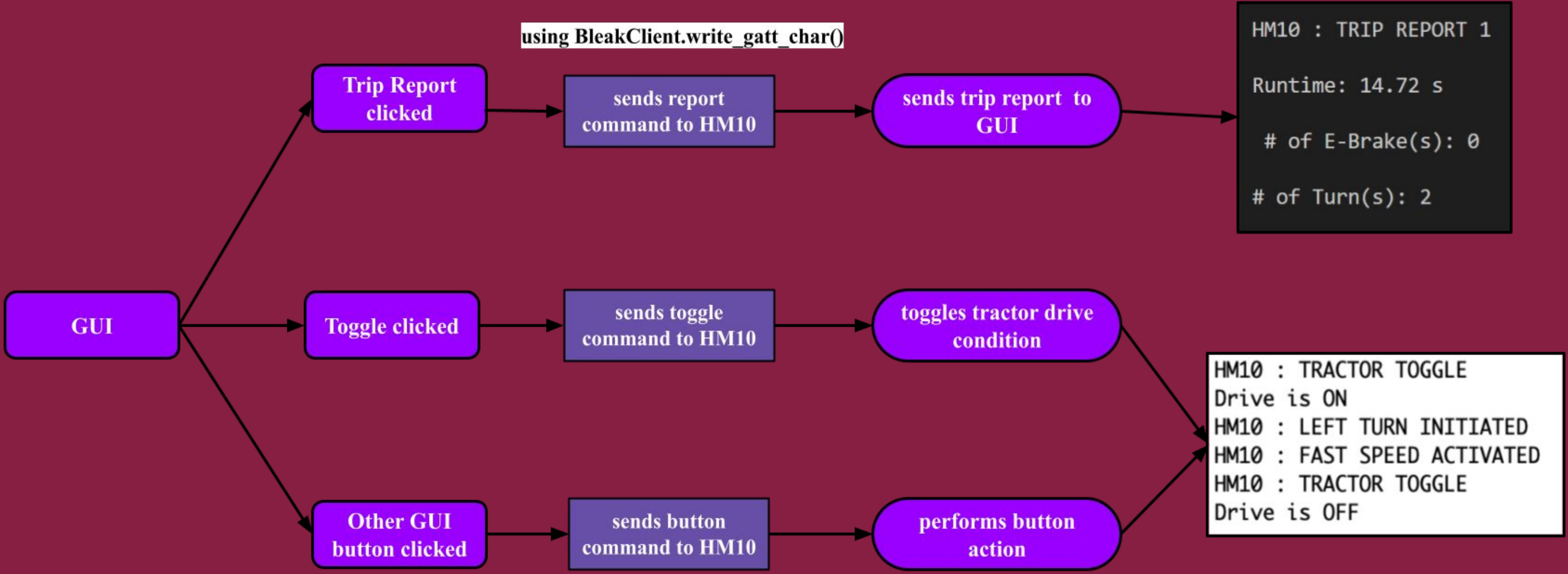
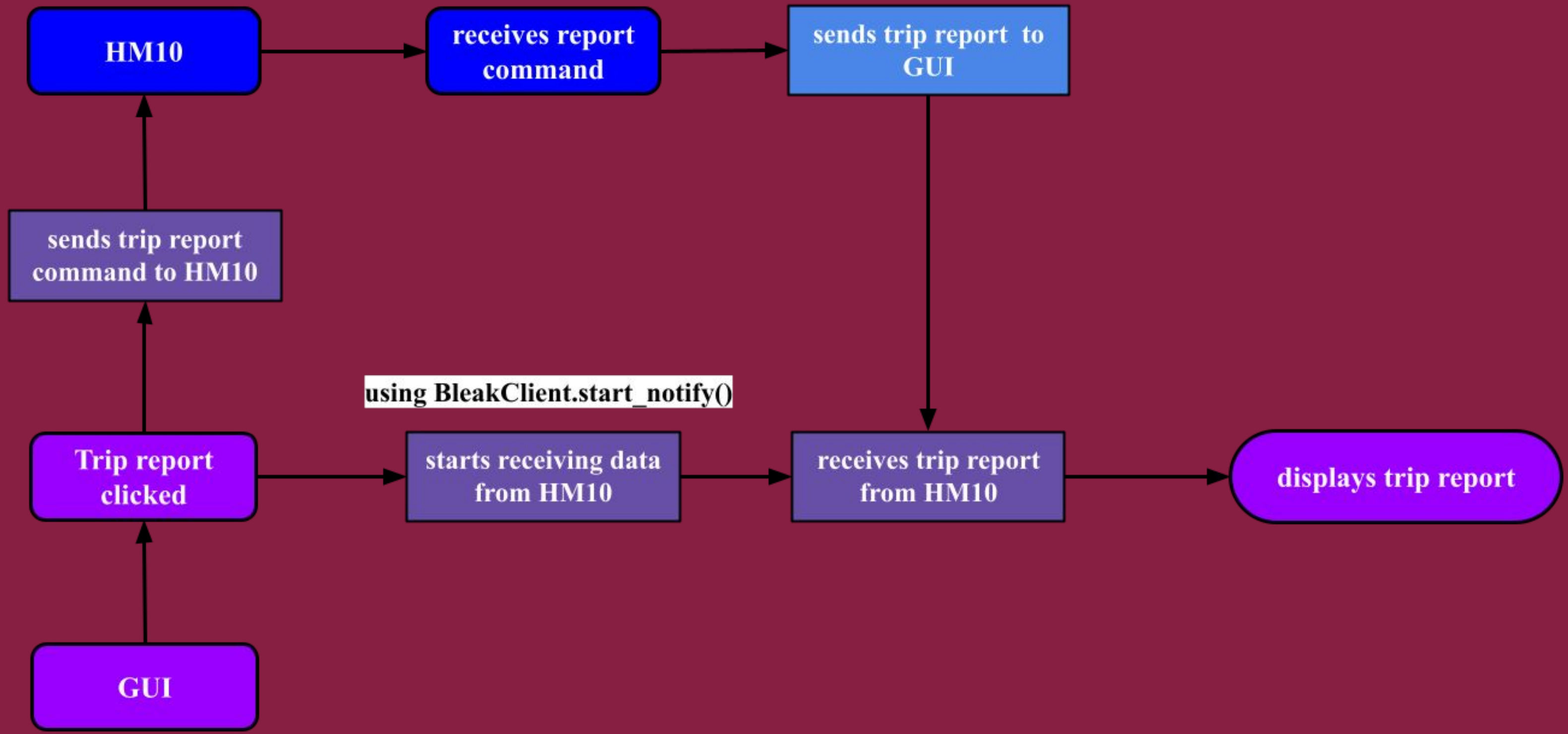
A deeper look…



Sending data to GUI



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Any Questions?

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